CASE REPORT

First study of safety and tolerability of 3,4-methylenedioxymethamphetamine (MDMA)-assisted psychotherapy in patients with alcohol use disorder: preliminary data on the first four participants

Ben Sessa,[1] Chloe Sakal,[2] Steve O'Brien,[1] David Nutt[1]

[1]Neuropsychopharmacology, Department of Medicine, Imperial College London, London, UK
[2]Department of Medicine, Avon and Wiltshire Mental Health Partnership NHS Trust, Bath, UK

Correspondence to
Dr Ben Sessa,
bensessa@gmail.com

Accepted 28 June 2019

Check for updates

© BMJ Publishing Group Limited 2019. No commercial re-use. See rights and permissions. Published by BMJ.

To cite: Sessa B, Sakal C, O'Brien S, *et al*. *BMJ Case Rep* 2019;**12**:e230109. doi:10.1136/bcr-2019-230109

SUMMARY

We present the preliminary data in an ongoing open-label safety and tolerability proof of concept study exploring the potential role for 3,4-methylenedioxymethamphetamine (MDMA)-assisted psychotherapy in treating patients with alcohol use disorder. At this stage, seven participants have completed the full 8-week MDMA-assisted psychotherapy course, including two therapy sessions each with MDMA. This paper focuses on the safety and tolerability of the therapeutic course for the first four participants to complete treatment. Longer-term outcomes of drinking behaviour will be presented later when the full project data are published. Results show all four participants have successfully tolerated the treatment. There have been no serious adverse events related to MDMA, no unexpected physiological responses to the MDMA sessions or changes to blood results or electrocardiograms, measured before and after the 8-week course. We conclude that the treatment is well-tolerated and are making plans to expand the project into a randomised placebo-controlled study.

BACKGROUND

3,4-methylenedioxymethamphetamine (MDMA) therapy has been studied extensively as a treatment for post-traumatic stress disorder (PTSD) in recent years, but there has never been a study to explore MDMA's role in treating addictions.[1] In recently published studies, the 'classic' psychedelic drugs have shown impressive safety and efficacy as potential tools to treat a range of different addictions.[2] MDMA-assisted psychotherapy has qualities that could also make it well suited to this population of patients.[3]

A quarter of adults in England consume alcohol harmfully, and around 6% of men and 2% of women are diagnosed as alcohol dependent.[4] Patients dependent on alcohol often have a history of psychological trauma and present with high levels of depression and social anxiety.[5] Alcohol-related health disorders, crime, accidents and domestic problems cost around £20 billion a year in England,[6] and in 2017, there were over 9000 alcohol-related deaths.[7]

Despite the massive burden of alcohol use disorder (AUD) in the UK, there remains a lack of consistency about the efficacy of alcohol treatments. Outcomes are generally poor, with rates of relapse after treatment up to 60% at 12 months[8] and 80% at 3 years.[9]

CASE PRESENTATIONS

Participant 1

A 34-year-old man with a 10-year history of AUD. He has attempted to stop drinking without formal services support on many occasions but always relapsed. He has no medical problems other than asymptomatic asthma, has no prior psychiatric history and is not on any medication. He has no history of trauma or abuse, lives in stable accommodation, is married with two children and is in full-time employment. There is no family history of substance misuse or mental illness.

Participant 2

A 63-year-old man with a 30-year history of AUD. He has never attempted recovery in the past. He has no medical problems, no prior psychiatric issues and is not on any medications. He has no history of trauma or abuse, is married with no children and worked full time before recently retiring. There is no family history of substance misuse or mental illness.

Participant 3

A 54-year-old woman with a 20-year history of AUD. She has had one previous inpatient detox and subsequently relapsed. She has no physical medical problems. She has experienced depression since adolescence and has made suicide attempts as a teenager and in the last 12 months. She is prescribed Quetiapine, Citalopram and Zopiclone. She described a difficult family life, with memories of neglect as a child, bullying at school and several unstable relationships as an adult. She has three children. There is no family history of substance misuse or mental illness.

Participant 4

A 50-year-old man with a 30-year history of AUD. He has attempted to stop drinking several times

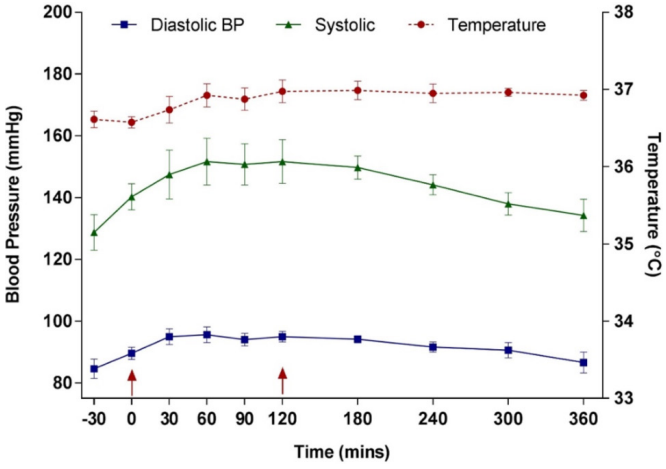


Figure 1 Blood pressure (BP) and temperature following MDMA at T=0, 125 mg (arrow) and T=120, 62.5 mg (arrow), n=4. Mean data combined for both MDMA therapy sessions, error bars±standard error of mean. MDMA, 3,4-methylenedioxymethamphetamine.

in the past without formal support but always relapsed within weeks. He has mild asthma and uses inhalers occasionally. He has no psychiatric history. He describes childhood memories of parental disharmony. He was frequently truant from school and started drinking from age 15 years and experimented with d-Lysergic diethylamide, psilocybin and ecstasy. He was a heroin user in his 20s, before successful treatment with methadone and has been opiate free for the last 20 years. There is no family history of psychiatric disorder or substance misuse.

INVESTIGATIONS

All four participants had normal ECGs and normal blood (urea and electrolytes, liver function tests and full blood count) at baseline assessment and 8 weeks, after receiving two experimental sessions with MDMA.

DIFFERENTIAL DIAGNOSIS

The diagnosis of AUD in each patient was confirmed by completion of the Structured Clinical Interview for the Diagnostic and Statistical Manual of Mental Disorders.[10]

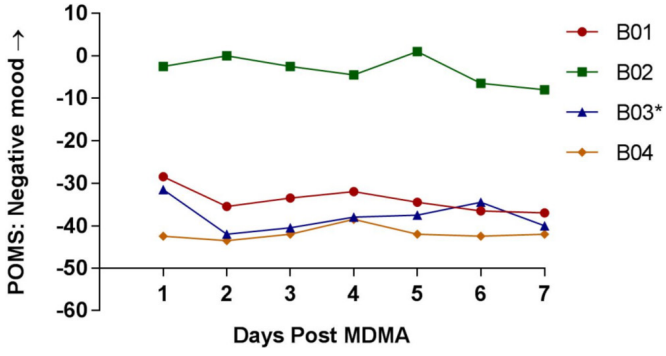


Figure 2 Individual POMS composite scores (n=4) for 7 days following MDMA-assisted psychotherapy (day 0). Data are mean of the two MDMA sessions. *On day 4, B03 scored very negatively due to an argument that was agreed unrelated to the study. The score for that day was subsequently removed. MDMA, 3,4-methylenedioxymethamphetamine; POMS, Profile of Moods States.

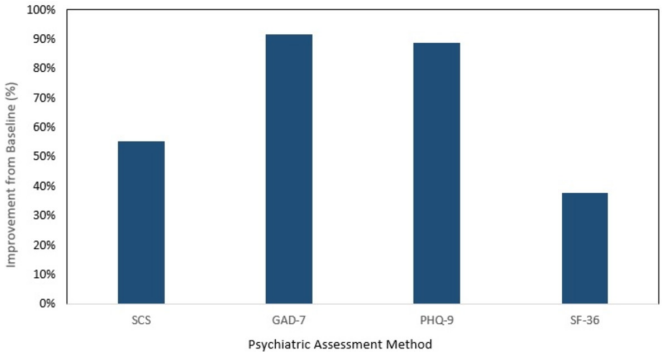


Figure 3 Self Compassion Scale (SCS), Generalised Health Questionnaire-7 (GAD-7), Patient Health Questionnaire-9 (PHQ-9) and Short Form Health Survey (SF-36) % improvement between baseline session and session 10, after completion of the 8-week psychotherapy course. Bars are mean data n=4.

TREATMENT

The ongoing MDMA-assisted psychotherapy study aims to recruit 20 adult patients with a primary diagnosis of AUD who have successfully undergone community alcohol detoxification. Exclusion criteria include a history of psychosis, personality disorder, serious suicidal risk, cardiac disease, severe liver disease, unstable hypertension, dependence on drugs other than alcohol, regular user of 'ecstasy', pregnancy and breastfeeding.

Eligible patients receive an 8-week course of MDMA-assisted psychotherapy, comprising 10 therapeutic sessions. The psychotherapy sessions were delivered by a male-female dyadic pair (typical for non-ordinary state of consciousness-assisted psychotherapy) consisting of a consultant psychiatrist and a senior clinical psychologist; both experienced in the fields of psychotherapy and addictions. The 1-hour weekly non-drug psychotherapy sessions employ aspects of Motivational Enhancement Therapy and are conducted on an outpatient basis. The participant was brought to the outpatient clinic by taxi during working hours and returned home after the 1-hour session by taxi.

On two of the sessions (weeks 3 and 6), patients are administered open-label MDMA (125 mg initially, followed by an optional 62.5 mg booster dose after 2 hours). The MDMA sessions last for 6–8 hours, and patients are monitored overnight at the treatment facility. The drug-assisted therapy sessions start at 10 am, and the drug effects wear off by 5 pm. At which point, the two therapists handed over care of the participant to two trained over-night medical attendants, who stayed in the treatment facility with the participant overnight. Participants are discharged home the following day, then telephoned daily for a week to collect data regarding quality of sleep, affect and potential suicide risk.

At the end of the 8-week therapeutic course, patients are followed- up for 9 months to assess longer-term data, including drinking behaviour.

Because patients with a history of heavy drinking frequently present with asymptomatic hypertension,[11] and acute MDMA is known to cause a transient rise in blood pressure and temperature, we monitor physiological data regularly throughout the MDMA-assisted psychotherapy sessions.

The study has full ethical approval from the Central Bristol NHS Research Ethics Committee; reference number 16/SW/0273 and approval from the Medicines and Healthcare products Regulatory Agency.

All participants were recruited from a local substance misuse service. Participation was voluntary, and all participants signed written consent forms to take part in the study and to consent for their data to be included in this British Medical Journal article.

OUTCOME AND FOLLOW-UP

Sixteen people have been screened so far for the study. Seven eligible participants have completed the 8-week open-label MDMA-assisted psychotherapy course and are still in the follow-up stage. Of the four patients included in this case series, two have remained totally alcohol free, and two have had a single one-off episode of low-dose alcohol use since finishing the therapeutic course. None of the four participants have returned to daily or harmful drinking during the therapeutic course. These results compare favourably with the pattern of heavy alcohol use, and all participants were engaged in before starting the study. There have been no serious adverse events related to the MDMA. During the MDMA sessions, none of the participants experienced unexpected abnormal physiological events requiring intervention. Figure 1 indicates while blood pressures rose gently and then declined back to baseline, as expected, all data remain within the normal range for people with a history of AUD.

No participants experienced any suicidality or sleep disturbance for 7 days after each of their MDMA sessions. One participant reported minor tinnitus and mild memory disturbance in the days after the MDMA sessions. However, these can be related to chronic postwithdrawal phenomena and not commonly related to MDMA. Mood was monitored in all participants for a week after each MDMA session using the Profile of Moods States questionnaire. Results demonstrate a consistent positively felt mood across all subjects for a week after each MDMA session. This suggests, contrary to some recreational ecstasy users' description of low mood in the days following use, there may, by contrast, be a potential 'afterglow' effect from clinical use of MDMA in a therapeutic environment (figure 2).

No participants reported any adverse psychological effects from the study interventions. Measurements of anxiety (Generalised Anxiety Disorder 7), depression (Patient Health Questionnaire 9), quality of life/functioning (Short-Form Health Survey) and compassion (Self-Compassion Score) were taken at baseline and week- 8 and at the end of the MDMA therapy course. All these measures improved during the course of the 8-week therapy (figure 3).

No participants have used recreational MDMA or ecstasy since taking part in the study, nor have they expressed any desire or craving to do so. There have been no drop-outs due to intolerance to the course of therapy. We are not reporting detailed outcome data in respect of drinking behaviour with this case series report, as the study is still ongoing.

DISCUSSION

There are no previous studies exploring MDMA-assisted psychotherapy as a treatment for AUD with which to compare this current ongoing study. However, MDMA Therapy—mainly for PTSD— has been explored since the 1980s.[12 13] Recent MDMA-PTSD studies have found statistically and clinically significant gains in symptom relief with no subjects reporting harm from participation.[14–16]

Carrying out clinical research on patients with AUD is difficult because of the high incidence of comorbidities in this population. Physical problems include alcohol-induced hypertension and impaired liver functioning. Psychological comorbidities include high levels of trauma, polysubstance drug misuse and personality disorders. Unstable social factors, such as unemployment, insecure housing and poverty, are also common in this group of patients—all of which added to the complexity of this project. Despite these challenges, the massive personal, clinical, societal and financial burden of AUD makes this diagnosis an important target for innovative research. This is especially imperative, given the current poor rates of lasting recovery with current best-approved treatments.

Patient's perspective

Participant 1: It's not about the drinking, the MDMA healed me inside and the drinking looks after itself … I'm seeing things anew, nature for the first time … I'm in control of my decisions, I've got control back … Life is just good!

Participant 2: I'm pleased I've managed to get through both the MDMA sessions … The overall package (of the MDMA therapy course) has been put together well … I went into it expecting to feel strange … It's called Ecstasy, but it was not an ecstatic feeling … (This course) is definitely a work in progress … I'm able to identify better when I am dealing with my feelings and when I'm doing things well … Its going to take quite a while to get fully better … I don't know how much of the changes I've made are due to the MDMA or due to the (non-drug) psychotherapy sessions.

Participant 3: Better than other treatments, including inpatient detox … I enjoyed every moment of it. Thrilled to be part of the study … I feel energised … The treatment has worked for me, done me a lot of good. I've got a lot of confidence out of it. I'm calmer … It's given me what I wanted; to be cured, to not have the cravings, to look at life differently. I'm not so angry at everything … Being under MDMA was beautiful. It showed me the real me; the me without alcohol.

Participant 4: A weight has been lifted off my shoulders. I haven't felt like that for a long time. There are no nagging doubts. I'm getting my life back on track … Everything is so much clearer. It's like a smog has been removed. I can see myself moving forward … It makes me think: why was I drinking that rubbish? I was just being stupid, idiotic, killing myself. There's no reason to be doing that … Taking part in this study has helped me focus more on life and my goals … An uplifting experience that I would recommend to anyone.

Learning points

- An 8-week course of 3,4-methylenedioxymethamphetamine (MDMA)-assisted psychotherapy can be safely delivered to this population of patients with alcohol use disorder (AUD).
- Patients with AUD tolerate the medicine and the psychotherapy course well, with no acute or lasting negative changes in physiological data up to 8 weeks.
- Demonstrable improvements in quality of life, mindfulness, self-compassion, anxiety and depression scores were observed in all participants following the 8-week course of MDMA therapy.
- This study forms the basis of further plans for a randomised double-blind placebo-controlled study testing the efficacy of MDMA-assisted psychotherapy as a treatment for AUD.

Acknowledgements The authors wish to thank the other team members of the BIMA study, including clinical psychologist and co-therapist Dr Laurie Higbed; Research Associates Dr Claire Durant and Dr Sue Wilson; and expert service user Kevin Speller. They are grateful to The Alexander Mosley Foundation for their funding support and the Multidisciplinary Association for Psychedelic Studies for academic support with the ongoing study. They are grateful to the local substance misuse service, Addaction, and Tim Williams for his support locally. In addition, they are thankful to the Bristol University CRIC Centre for hosting the study and Imperial College London for their academic sponsorship.

Contributors BS was the principal investigator for study, corresponding author, and main contributor to the writing of the paper. CS contributed to data entry and creating graph for paper. SO'B was a research assistant and contributed to data entry and creating graphs for paper. DN was co-investigator for study and approved the paper.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Obtained.

Provenance and peer review Not commissioned; externally peer reviewed.

REFERENCES

1 Jerome L, Schuster S, Yazar-Klosinski BB. Can MDMA play a role in the treatment of substance abuse? *Curr Drug Abuse Rev* 2013;6:54–62.
2 Sessa B, Johnson MW. Can psychedelic compounds play a part in drug dependence therapy? *Br J Psychiatry* 2015;206:1–3.
3 Sessa B. Why MDMA therapy for alcohol use disorder? And why now? *Neuropharmacology* 2018;142:83–8.
4 NICE. Alcohol-use disorders: diagnosis, assessment and management of harmful drinking and alcohol dependence. NICE clinical guideline 115. 2011 www.nice.org.uk/CG115
5 Stewart SH. Alcohol abuse in individuals exposed to trauma: a critical review. *Psychol Bull* 1996;120:83–112.
6 HM Government. *The Government's Alcohol Strategy*: Secretary of State for the Home Department HM Government, 2012.
7 Office for National Statistics. Alcohol-specific deaths in the UK: registered in 2017. https://www.ons.gov.uk/peoplepopulationandcommunity/healthandsocialcare/causesofdeath/bulletins/alcoholrelateddeathsintheunitedkingdom/registeredin2017 (Accessed 26th Feb 2019).
8 Evren C, Durkaya M, Dalbudak E, *et al*. Factors related with relapse in male alcohol dependents: 12 months follow-up study. *J Psychiatry Neurol Sci* 2010;23:92–9.
9 Moos RH, Moos BS. Rates and predictors of relapse after natural and treated remission from alcohol use disorders. *Addiction* 2006;101:212–22.
10 First MB, Williams JBW, Karg RS, *et al*. *Structured clinical interview for DSM-5—research version (SCID-5 for DSM-5, Research Version; SCID-5-RV)*. Arlington, VA: American Psychiatric Association, 2015.
11 Tsuruta M, Adachi H, Hirai Y, *et al*. Association between alcohol intake and development of hypertension in Japanese normotensive men: 12-year follow-up study. *Am J Hypertens* 2000;13:482–7.
12 Greer G, Tolbert R. Subjective reports of the effects of MDMA in a clinical setting. *J Psychoactive Drugs* 1986;18:319–27.
13 Sessa B. Could MDMA be useful in the treatment of PTSD? *Progress in Neurol & Psychiatry* 2012;15:4–7.
14 Mithoefer MC, Wagner MT, Mithoefer AT, *et al*. The safety and efficacy of {+/-}3,4-methylenedioxymethamphetamine-assisted psychotherapy in subjects with chronic, treatment-resistant posttraumatic stress disorder: the first randomized controlled pilot study. *J Psychopharmacol* 2011;25:439–52.
15 Mithoefer MC, Wagner MT, Mithoefer AT, *et al*. Durability of improvement in post-traumatic stress disorder symptoms and absence of harmful effects or drug dependency after 3,4-methylenedioxymethamphetamine-assisted psychotherapy: a prospective long-term follow-up study. *J Psychopharmacol* 2013;27:28–39.
16 Oehen P, Traber R, Widmer V, *et al*. Pilot study of MDMA-assisted psychotherapy for treatment-resistant PTSD. *J Psychopharmacol* 2013;27:40–52.

Copyright 2019 BMJ Publishing Group. All rights reserved. For permission to reuse any of this content visit
https://www.bmj.com/company/products-services/rights-and-licensing/permissions/
BMJ Case Report Fellows may re-use this article for personal use and teaching without any further permission.

Become a Fellow of BMJ Case Reports today and you can:
- ► Submit as many cases as you like
- ► Enjoy fast sympathetic peer review and rapid publication of accepted articles
- ► Access all the published articles
- ► Re-use any of the published material for personal use and teaching without further permission

Customer Service
If you have any further queries about your subscription, please contact our customer services team on +44 (0) 207111 1105 or via email at support@bmj.com.

Visit casereports.bmj.com for more articles like this and to become a Fellow